LEF MODEL · CHAPTER 25

Chapter 25: The Fundamental Requirements of the LEF Model (R1–R8) — Isotopes, the Core, the Crust, the Centre-of-Mass Offset and the Exposure Zones

Based on Paper 13 — 13. The Requirements: What the mechanism forces, what the Moon has, and what remains unresolved (Parts A–B and Appendix A: requirements R1–R8, the mass balances and the statistical tests T1–T5)

01 · THE MYSTERY

The mystery and the methodology: separating the formation environment from the modern Moon, and ruling out a "cumulative score"

After building, in the previous chapters, the mechanics of the Earth's pear, the orbit and the laws of moons and rings, **Paper 13 (The Requirements)** puts the whole model to a strict methodological test. To prevent statistical bias or a mixing of assumptions with measurements, the model sets four ground rules:

1. **A methodological partition between the formation environment and the modern Moon:** the formation environment is a soft, hot, fast-spinning system (T = 5.5–5.77 hours, initial mass $M_0 = M_E + M_L = 6.0455 \times 10^{24}$ kg, equatorial radius 6,721 km); while the modern Moon is a frozen evolutionary product that went through billions of years of cooling and late volcanic

activity. A late measurement may not be used as an initial condition without a physical model linking the two.

2. **A threefold structure for each requirement, and ruling out a "cumulative score":** each requirement is defined by (a) the mechanical condition the model requires, (b) the quantitative measure that tests it, and (c) the result that would count as a mismatch. The model **forbids adding successes into a weighted score**, both because crustal thickness, the centre-of-mass offset and the maria are different expressions of *one and the same mass distribution* (and adding them would count one fact twice), and so as not to hide a mismatch in one requirement behind successes in others.

3. **A chronological distinction between a derived result, a postdiction and a prediction:** the paper strictly separates a *derived result* (following directly from the equations), a *postdiction* matching existing observations in the literature, and an independent *prediction* not yet tested.

02 · AN IMAGE

An image: the thickening helmet, the peeling suction cup, and the bottle sealed in lava

Four tangible images illustrate the eight fundamental requirements (R1–R8) and the mass balances behind them:

- **The helmet that thickens threefold (R3 — the asymmetry of the crust):** the Earth's dome that rose on the head of the pear carried an ancient lithospheric envelope 18 km thick (density 2,800 kg/m^3) over a hemisphere with a 6,000 km base (total mass 2.85×10^{21} kg, about 3.9% of the Moon's mass). When the dome rounded and closed over the far hemisphere, three times smaller in area, conservation of mass forced a threefold linear thickening — to **50–60 km** (predicted mass 2.6×10^{21} kg at 55 km, a match of about 10% with no tuning!).
- **The suction cup that peels from the edges inward (R8 — the three exposure zones):** while the area of the dome (the far side) was exposed to space from the start, the base sat pressed against the Earth. During the rounding, the

base lifted and peeled from its edges to its centre like a suction cup coming off glass — so the central contact dome was exposed to space *last*, right at the moment of detachment at the corotation line (16,399 km).

- **Two dancers holding hands (R4 — synchronous locking from birth):** a body mechanically attached to the tip of the pear completes exactly one rotation on its axis for every orbit ($\omega_{\mathrm{rot}} = \omega_{\mathrm{orb}}$), so its attached side always faces the parent even before the moment of detachment.
- **The bottle sealed inside an olivine crystal (R6 — the preservation of volatiles):** tiny droplets of magma (melt inclusions) trapped inside olivine crystals beneath the surface kept their original water content (615–1,410 ppm) safe from degassing in eruption, like a bottle sealed inside a rock.

03 · SCIENCE AND MODEL

Science and model: the first eight requirements (R1–R8) and the statistical tests (T1–T5)

Parts A and B of Paper 13 set each of the eight requirements (R1–R8) against the most up-to-date empirical data:

1. **R1 — closeness to a terrestrial source (oxygen isotopes):** precise laboratory measurements (Young et al. 2016) show that the oxygen-isotope gap between the Moon and the Earth is $\Delta^{17}\mathrm{O}_{\mathrm{L-E}} = -1 \pm 5$ ppm — a statistical zero within the measurement resolution, as required of a body built entirely from the Earth's reservoirs (deep mantle and the sial layer). The combined multi-system balance (O, Ti, Cr, W) is defined as an open task.

2. **R2 — a tiny metallic core as a transport target:** the Apollo seismograph network (Weber et al. 2011) measured a solid inner core with a radius of 240 km and a liquid outer core with a radius of 330 km — about **only 1.4% of the Moon's mass**. For a deep-material fraction $f_{\mathrm{int}} = 0.95$, the balance equation $f_{\mathrm{core}} = f_{\mathrm{int}} X_{\mathrm{met}} \epsilon_{\mathrm{core}}$ sets a sharp hydrodynamic target: $X_{\mathrm{met}} \epsilon_{\mathrm{core}} \simeq 1.47\%$.

3. **R3 and R7 — the asymmetry of the crust and the centre-of-mass offset:** the GRAIL gravity maps (Zuber et al. 2013) confirm that the crust on the far side is significantly thicker than on the near side (global average 34–43 km).

The near crust is the Earth's sial (SiAl) layer that peeled off the base of the structure (mass 1.4–1.7 × 10^{21} kg). Calculating the moment integral of an asymmetric crust (Appendix A) yields a centre-of-mass offset toward the near side in the range of **1.24–3.11 km** for density contrasts of $\Delta\rho = 300–750$ kg/m³ — containing the value measured on the Moon (**~ 2 km**, and coinciding with it exactly at $\Delta\rho \simeq 480$ kg/m³).

4. **R5 — separating the four geochronological clocks:** the paper forbids comparing the age of a late basalt (clock 3 — such as the Chang'e-5 samples at 2.0 Ga on the near side and Chang'e-6 at 2.8 Ga on the far side) with the crystallisation age of the primary crust (clock 1). The order of stabilisation must be measured only by comparing primary crust units of the same genetic type from both sides.

5. **R8 — the three concentric exposure zones and running tests T1–T5:** around the centre of the near side (in the updated geometry around the contact centre at 24.1°N, 335.2°E — an angular deviation of $\gamma = 34.0^\circ$ from the sub-Earth point 0°, 0°), the Moon's surface is arranged in three concentric belts: the contact dome (0°–31°, the thinnest crust and strong mascons without negative rings), the edge belt (31°–70°, mascon basins with a negative ring, such as Orientale), and the far side (70°–180°, thick crust and basins without mascons).

04 · THE MECHANISM IN WORDS

The mechanism in words: the mass balance of the dome, the helmet model and deciding the exposure zones against the PKT

Parts B.3, B.7–B.8 and Appendix A of Paper 13 show how the geometry of the Moon's rounding joins crustal thickness, the centre-of-mass offset and the belt of basins into one closed system of equations:

1. **A critical distinction between end values and hemispheric averages (§B.3):** in the long-wavelength helmet model $h(\psi) = h_0 - A\cos\psi$, point end values of 30 km on the contact axis ($\psi = 0^\circ$) and 60 km on the far side

($\psi = 180^\circ$) give a baseline of $h_0 = 45$ km and an amplitude A = 15 km. Since the area-weighted average of $\cos\psi$ over a hemisphere equals 0.5, the hemispheric averages of this example are **37.5 km and 52.5 km** (and not 30 and 60 km!).

2. **The separate mass balance of the far side and the near side (§B.3):** the Earth's hemispheric dome ($h_s = 18$ km, $\rho_s = 2,800$ kg/m³, base radius $R_h = 3,000$ km) carries a mass of **2.85 × 10²¹ kg** (3.9% of the Moon's mass). As it closes over the far hemisphere, three times smaller in area, it supplies, to within about 10%, the mass of the far crust at a thickness of 55 km (**2.6 × 10²¹ kg**, 3.5% of the Moon's mass); while the near crust (30–35 km, mass 1.4–1.7 × 10²¹ kg) is the Earth's sial layer that peeled off the base of the structure.

3. **A structural decision against the PKT thermal-anomaly hypothesis (§B.8):** alternative models explain the Moon's two-sidedness by a one-sided thermal anomaly (the Procellarum KREEP Terrane — a hot near side and a cold far side), or by the radiation of an ancient "Earthshine" (Roy et al. 2014). The LEF model predicts **a concentric three-zone structure** that includes *an intermediate edge belt* (31°–70°) in which **mascon basins with a negative ring** are selectively concentrated (such as Orientale on the western edge and Smythii on the eastern), because the edges of the base dome were released gradually from all sides.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The equation of concentric crustal thickness, the mass-moment integral for the centre-of-mass offset for a layer of finite thickness $b = R_L - h_0$ (Appendix A), and the calculation of the spatial deviation angle γ of the contact centre are formulated as follows (§B.3, §B.7, Appendix A):

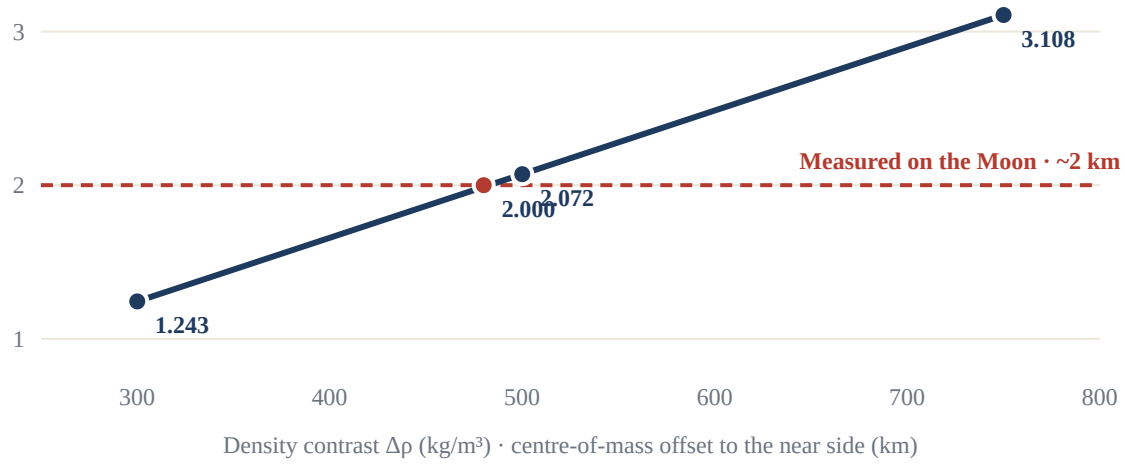
$$h(\psi) = h_0 - A\cos\psi$$

$$M_{\text{shell}} = \rho_s h_s (2\pi R_h^2) = 2.85 \times 10^{21}\ \text{kg}$$

$$\delta x = \frac{4\pi\Delta\rho}{3M_L}\left(b^3 A + \frac{3}{5}bA^3\right) \simeq \frac{\Delta\rho}{\bar{\rho}_L}A$$

$$\cos\gamma = \cos(24.1^\circ)\cos(24.8^\circ) \implies \gamma = 34.0^\circ$$

In the moment integral $M_L \delta x = \frac{\pi\Delta\rho}{2}\int_{-1}^{1} u(b+Au)^4\,du$ all the terms with an odd power of $u = \cos\psi$ vanish, so only the asymmetric component (A) contributes to shifting the centre of mass toward the near side (+2.072 km for $\Delta\rho = 500$ kg/m³), while every symmetric spherical layer (h_0, the mantle and the core) cancels out completely.



The centre-of-mass offset from the finite-thickness integration (Appendix A): the range 1.24–3.11 km contains the measured value, and coincides with it at Δρ ≈ 480.

06 · THE SIGNATURE

The signature in reality: the results of running tests T1, T2 and T5, and calibrating the centre-of-mass offset

Running the scripts on the GRAIL data and on the Robbins crater catalogue (≥ 20 km) yields a transparent and precise picture of the state of the first eight requirements (§B.7–B.9):

- **A clear success in the axis and crater-density tests (T5 and T2):** the Spearman correlation $r_s = +0.87$ (p = 2 × 10^{-4}) around the sub-Earth axis exceeds every other control axis (the eastern edge +0.76, Procellarum +0.74, the south pole +0.43, the north −0.43), and the density of large craters on the far side is 2.21 times higher than in the contact dome (and in the updated geometry 8.42 times including the maria, and 4.15 times without them).
- **A transparent record of the T1 threshold and continued testing of T3–T4:** in test T1, the monotonic order holds in all four GRAIL models, but the difference in model 1 (9.65 km and 9.9 km) is 0.1–0.35 km below the lower threshold (10 km) and so is recorded as *did not pass*, while the basin tests T3 and T4 are set for a full quantitative run in the reproduction package.

The lab box: summary of the eight fundamental requirements (R1–R8), the exposure-zone tests (T1–T5) and the sensitivity of the centre-of-mass offset (Paper 13, Parts A–B and Appendix A)

Requirement	Subject and the physical requirement	The empirical finding measured today	Status and the open task
R1	**Closeness to a terrestrial source (mixture of mantle + sial)**	**Identity in oxygen isotopes:** $\Delta^{17}\mathrm{O}_{\mathrm{L-E}} = -1 \pm 5$ ppm	**Matches (open: multi-system balance O, Ti, Cr, W)**
R2	**A tiny metallic core ($X_{\mathrm{met}}\epsilon_{\mathrm{core}} \simeq 1.47\%$)**	**A seismic core with a radius of 330 km (≈ 1.4% of the Moon's mass)**	**Matches (open: a hydrodynamic model for transporting iron droplets)**
R3	**Crustal asymmetry and the mass balance of the dome (18 km)**	**A thick far crust; balance 2.85 against 2.6 × 10^{21} kg (~ 10%)**	**Matches (open: a full spatial profile and the source of the near sial)**
R4	Synchronous locking from the moment of detachment at the corotation line	1:1 synchronous locking with the near side facing the Earth	**Matches** (open: continuity of the orientation vector in the orbit)
R5	Order of thermal stabilisation (far side before near side)	Separate late basalts (2.0 Ga and 2.8 Ga, clock 3)	**Matches** (open: comparing clock 1 in primary highland crust)

R6	**Presence of volatiles to lower viscosity and density**	**Water in melt inclusions in olivine: 615–1,410 ppm**	**Matches (open: a thermodynamic model of fluxes and phases)**
R7	**A quantitative link between the crust and the centre-of-mass offset**	**Measured offset ~ 2 km to the near side; model range 1.24–3.11 km**	**Matches (open: a joint vector inversion)**
R8	Three concentric exposure zones (0°–31°–70°–180°)	**T5 passed** ($r_s = +0.87$), **T2 passed**, **T1 did not pass** (9.65–9.9 km)	**Open for completion** (basin tests T3 and T4 and sorting of crater ages)

Density contrast $\Delta\rho$ (kg/m³)	Centre-of-mass offset in the thin-shell approximation (km)	Centre-of-mass offset in the finite-thickness integration (km)	Match to the geodetic observation (~ 2 km)
300	1.345	1.243	Lower bound of the sensitivity range
500 (~ 480 when calibrated)	**2.242**	**2.072 (2.000 at 480)**	**An exact match to the measured order of magnitude (2 km)**
750	3.362	3.108	Upper bound of the sensitivity range

07 · NEXT

The door ahead

In this chapter we surveyed the methodological framework of Paper 13 and the eight fundamental requirements (**R1–R8**) — from the isotopic identity (−1 ± 5 ppm), through the tiny core (1.4%), the crustal balance (2.85 against 2.6 × 10^{21} kg) and the centre-of-mass offset (2.07 km), to running the statistical tests T1, T2 and T5 on the concentric exposure zones.

In the next chapter — **Chapter 26: The Ten Further Requirements (R9–R18), the Hafnium–Tungsten Clock and the Map of Future LEF Research** (from Paper 13 — Parts C–D and Appendix B) — we will complete the whole system: we will examine the 7 requirements of the pear (R9–R15: the LLVP lower anomaly in the Pacific, the KREEP complex, the absence of foreign material and the compaction of craters at

the poles), the 3 requirements after the pear (R16–R18), the constraint of the radioactive-decay clock $^{182}\mathrm{Hf}$–$^{182}\mathrm{W}$ over a 600-million-year window, and the 7 open tasks for researchers!

Chapter 26

The Ten Further Requirements (R9–R18), the Hafnium–Tungsten Clock and the Map of Future LEF Research →

← Chapter 24 · Prologue

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